Motor Ontology

Why does the world move?

Time as a counter, reality as frames

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Contents

Introduction 1

I Doubt 2

1 What puzzles us about time 3

2 The problem of the river and the observer 5

3 What does it mean to 'be'? 7

II The Model 9

4 The frame structure of reality 10

5 Dimensions and fields: who is who 12

6 $n = 0$ as null, not as 'zero particles' 14

7 Consciousness as external input 15

8 Synchronisation of dimensions: why everything happens at one speed 16

9 Consciousness and free will – the conclusion 17

III Consequences and Reflections 18

10 The motor ontology and modern physics 19

11 Unanswered questions 20

12 The human being in this picture of the world 21

Epilogue 22

Introduction

"Time is not a thing, it is an accounting of motion."

—— Author

When we look at the world, we see movement. Stars rotate, rivers flow, people walk, thoughts succeed one another. Everything around us is a process. But what makes this process possible? Why does the world not freeze in one state? Why do we feel that time passes and does not stand still?

Physics describes the laws of motion, but does not explain *why* that motion occurs. It says: 'There is time', but does not answer what time really is. It says: 'There are laws', but does not specify where they are written or how they are executed.

This book offers a different approach. We will not delve into formulas. We will reason logically, relying on simple analogies. We will try to see behind the familiar world – a world of frames, flows, and dimensions.

This is not a physics textbook. This is an invitation to reflect.

Part I

Doubt

Chapter 1

What puzzles us about time

> "The only reason for time is so that everything doesn't happen at once."
>
> —— Albert Einstein (attributed)

When we look at a clock, we think we see time. But in fact we see a hand that moves across the dial, or digits that change on a screen. We say: 'Five minutes have passed.' But what are 'five minutes'? They are 300 swings of a pendulum, or 300 turns of a gear, or 300 billion oscillations of a caesium atom.

Time is always a count of something else. We never measure time directly. We measure the number of repetitions of some process: the rotation of the Earth, the beating of a heart, the vibration of a crystal.

This is like measuring 'length' in metres, where metres are defined by the number of steps, and steps by the number of heartbeats. We would then think that 'length' is something real, when in fact we were counting steps.

It is the same with time: we are used to thinking of it as a substance, a 'river' along which we float. But what if there is no river? What if there is only a process that we call 'time', but the process itself is something else?

Imagine a three-dimensional object, say an apple. We can draw it, measure it, describe it. If we add a parameter t to its description, nothing changes. The apple remains motionless. It does not start falling, rotting, or changing colour.

Why? Because t is just a label, a number. For the apple to come alive, we need not a number, but a mechanism that recalculates its state: coordinates, bonds between atoms, interactions with fields. We need a process that creates a new apple in a new instant, slightly different from the previous one.

This process I call the motor dimension.

The motor dimension is not a place, not a fifth coordinate. It is an action. It is what turns a static snapshot of the world into a living, moving reality. It works like a film projector showing frames at enormous speed. Each frame is the complete configuration of the Universe at that moment. The next frame is the result of applying the laws of physics to the previous one.

And here arises the key insight: time is not a flow, but a frame counter. We

say ‘a second has passed’ – meaning the motor dimension has performed a certain number of update cycles. We say ‘time flows’ – meaning frames replace one another. But the counter itself does not move the world; the process moves it.

Chapter 2

The problem of the river and the observer

> "The distinction between past, present and future is only a stubbornly persistent illusion."
>
> —— Albert Einstein

In 20th-century physics, a picture of the world known as the block universe became established. Imagine a four-dimensional cube: three axes are space, the fourth is time. In this cube, all moments already exist – from the Big Bang to the heat death of the Universe. The past, present, and future are equally real. We simply float along this river, and our consciousness moves from one cross-section to another.

This picture is very beautiful, elegant, mathematically consistent. But it has one serious problem that is rarely discussed.

If all moments of time exist simultaneously, then where exactly are we now?

Imagine a river divided into 100 segments. Each segment is assigned a number – 1, 2, 3 ... 100. We are at some one of them. But why exactly at number 43, and not at 12 or 98? In the block universe, there is no reason for this moment to be 'our present'. It is just a given that we accept without explanation.

Moreover, if all states already exist, we could, in principle, be in any of them. Why do we not jump from one moment to another? Why do we sequentially experience 43rd, then 44th, then 45th, and not leap from 43rd straight to 100th?

This is like a calculator. If we type 1+1+1+...+1 (100 times) and press 'equals', the calculator instantly gives 100. It does not stop at intermediate numbers. Why does reality behave differently? Why are we not immediately in the final state of the Universe?

In my model, the answer to this question is simple and natural.

The future does not exist until it is computed.

The motor dimension is not merely 'showing an already finished film'. It is the process of creating the film right now. At each step, the processor takes the current frame, applies the laws of physics, takes input from consciousness, and creates the

next frame.

That is exactly why we are at frame 43 and not at frame 100. Frame 100 does not yet exist. We cannot be there because it has not yet been created. We go step by step because each next frame depends on the previous one, and also on decisions that have not yet been made.

Chapter 3

What does it mean to ‘be’?

> “To be is to be perceived.”
>
> —— George Berkeley (paraphrased)

We are used to thinking that the world consists of things. A table, a chair, an apple, a person – these are objects that have properties: mass, colour, shape. They exist independently of whether we look at them or not. This intuition is so strong that we transfer it to all of reality, even to atoms and fields.

But what is a ‘thing’ really?

Take an apple. We see it as red, feel its firmness, smell its fragrance. But if we look at it through a microscope, we see not a solid surface but a multitude of molecules with emptiness between them. If we look deeper – we see atoms consisting of protons, neutrons, and electrons that move continuously, oscillate, interact.

An apple is not an object. It is a process.

It consists of particles that move. Atoms vibrate, molecules exchange electrons, chemical reactions occur continuously. If we stop all this movement – if we freeze the apple to absolute zero and stop all quantum oscillations – it ceases to be an apple in the usual sense. It becomes just a set of particles in fixed positions, a dead structure.

The same applies to a person. We are not just a set of cells. We are a flow. Our cells are renewed, our thoughts change, our consciousness continuously processes information. We exist only because this flow is not interrupted.

To be is to participate in a process.

Now let us ask ourselves: what makes this process possible? What makes particles move, atoms vibrate, thoughts succeed one another?

In standard physics, there is no answer to this question. Physics describes motion, but does not explain why it exists. It says: ‘There are laws that determine the evolution of a system.’ But laws themselves are merely rules written on paper. Where are they written in reality? How do particles ‘know’ these laws?

In the motor ontology, the answer is simple and clear.

Motion is not a property of objects. Motion is the very way objects exist.

The motor dimension is the process that creates the next frame. An object exists

only as long as its data is carried over from one frame to the next. As soon as this transfer stops – the object falls out of the flow, and it is no longer in reality.

Part II

The Model

Chapter 4

The frame structure of reality

"Life is what happens when you're busy making other plans."

—— John Lennon (paraphrased)

So we have concluded: the world is not a set of things, but a continuous process. But what is this process? How exactly is it organised?

Imagine you are watching an old film on celluloid. You see movement, life, events. But if you stop the projector and look at the film, you see individual frames – motionless pictures that differ from each other by fractions of a millimetre. Movement arises only when these frames are quickly replaced.

The Universe works exactly the same way. But there are three important differences from cinema:

1. Frames in cinema are flat images. Frames of the Universe are complete three-dimensional configurations of everything. Each such frame contains all fields, all particles, all their properties: coordinates, masses, charges, spins, velocities. It is not a picture – it is a complete snapshot of reality.

2. In cinema, the frames are already shot and recorded. The film exists as a whole, even if we have not yet watched it. In our model, future frames do not exist in advance. They are created right now, before our eyes.

3. In cinema, the projector simply shows a ready-made film. In our model, the projector is not just a display, but a creation. It takes the current frame, applies the laws of physics, takes input from consciousness, and creates the next frame.

What is the motor dimension?

This is the most important term. It must be understood correctly.

The word 'dimension' is usually associated with space. But the motor dimension is not space. It is a process that creates a sequence of frames.

Imagine an escalator in the metro. The steps move, and a person standing on them is transported. The steps themselves are frames. The mechanism that moves the escalator – that is the motor dimension.

The motor dimension has no size. It is not a 'place' one can enter. It is an action that occurs everywhere and always. It permeates all frames, like light permeates a room. You cannot see the light itself – you only see what it illuminates.

In our model, the motor dimension works at a colossal frequency. Each cycle is a transition from one frame to the next. This frequency is so high that we do not notice the discreteness. To us, the world seems continuous, just as cinema seems continuous even though it is actually a sequence of frames.

Chapter 5

Dimensions and fields: who is who

"The map is not the territory."

—— Alfred Korzybski

In physics, we constantly hear the words 'field' and 'dimension'. These terms are used often, but their distinction remains blurred. In the motor ontology, we introduce a clear, operational distinction.

What is a field?

A field is data. It stores information about the state of objects in the current frame: their coordinates, masses, charges, spins, velocities. It is like the RAM of a computer.

A field is not active. It does not make decisions, does not compute, does not change itself. It simply exists inside the frame.

Examples of fields: electromagnetic field, Higgs field, coordinate field.

What is a dimension?

A dimension is an active entity. It is either the medium in which events occur, or the process that creates those events, or the rule by which they occur.

A dimension permeates all frames without exception.

Examples of dimensions: three-dimensional space (the stage), the motor dimension (the processor), the gravitational dimension (the rule of curvature), the informational dimension (consciousness).

The key distinction

The following table clearly shows how a field differs from a dimension.

Field	**Dimension**
Passive data	Active medium or process
Stores the state of objects	Sets the rules or performs computations
Completely recalculated in each frame	Continuously permeates all frames
Has no inherent dynamics	Is the source of dynamics
Example: electromagnetic field, Higgs field	Example: space, gravity, consciousness

Table 5.1: The distinction between field and dimension in the motor ontology

Important: every dimension can be described as a field, but not every field is a dimension. For instance, space can be described as a field of coordinates, but it is not recalculated – it is always present as a stage. And the electromagnetic field is data; it does not set rules.

Chapter 6

$n = 0$ as null, not as 'zero particles'

"Nothingness is not nothing."

—— Philip K. Dick (paraphrased)

In standard physics, zero usually means 'absence'. In our model, we introduce a different understanding. $n = 0$ is falling out of the flow.

Recall: the motor dimension is the process that transfers data from frame to frame. Each object has some 'momentum' along this process, which we denote by the number n.

- If $n \neq 0$, the object is 'hooked' onto the flow. Its data is carried over. - If $n = 0$, the object loses this momentum and stops being updated.

Analogy with a river and fruit: you throw an apple onto the bank – it stops floating, falls out of the flow. For you, it disappears.

What lies 'outside' the flow? We do not know, and cannot know, because all our observations are confined to frames.

$n = 0$ is not emptiness, but exclusion from reality.

Chapter 7

Consciousness as external input

> "I think, therefore I am."
>
> —— René Descartes

Now about the most important thing. Why do we feel that we are alive? Why are we not simply automata?

In the motor ontology, consciousness is not computation. It is external input.

Imagine a computer program simulating a billiards table. The balls move according to the laws of physics. But if the program has a player who can strike a ball at any moment, his actions cannot be computed in advance.

Conscious beings can change the trajectory of passive objects. They introduce new data into the system, which is not derivable from the previous state.

We postulate the existence of an informational dimension – a special dimension that stores and carries consciousness from frame to frame.

The informational dimension is the source of free will. Without it, the world would be a mechanical system.

Important: the motor dimension does not 'wait' for a command from consciousness in the sense of pausing. It works in synchrony with the informational dimension, which is part of the common rhythm. The content of consciousness (choice) is not computed, but is taken into account at each step as part of a synchronised process. More about this in the next chapter.

Chapter 8

Synchronisation of dimensions: why everything happens at one speed

> "All things come into being by conflict of opposites."
>
> —— Heraclitus

We have already said that reality is a single coordinated mechanism. All dimensions must work in the same tempo, otherwise chaos would ensue.

Imagine an orchestra. If each musician plays at his own tempo, it would be a cacophony. The conductor sets the common rhythm.

In our model, the conductor is the motor dimension. It sets the frequency of frame updates. All other dimensions (gravity, electromagnetism, strong and weak interactions, and the informational dimension) work synchronously in the same tempo.

If one dimension worked faster or slower, physical processes would become mismatched. A ball thrown into the air could first fly by inertia, then suddenly fall or jerk upward. Atoms would fall apart, light would cease to propagate predictably. But this does not happen, because the synchronisation is absolute.

The motor dimension does not 'dictate' speed, but is the common beat that holds everything together. It does not wait for a command from consciousness, but includes it as an integral part at each step. Consciousness is not an external factor, but an embedded element of the system – yet its content remains free.

That is exactly why the world is not predetermined, and exactly why we experience it as alive.

Chapter 9

Consciousness and free will – the conclusion

> "Man is condemned to be free."
>
> —— Jean-Paul Sartre

Now we can give a final answer to the question of free will.

Free will is not an illusion. It is real, because consciousness is not derived from data. It introduces something new that was not contained in the initial conditions.

The informational dimension does not compute the choice – it provides it. The motor dimension takes this choice into account at every step, synchronously with other laws.

We cannot control the laws of physics, but we can choose how to use them. Every choice we make changes the next frame. This gives us responsibility.

Part III

Consequences and Reflections

Chapter 10

The motor ontology and modern physics

> "The important thing is not to stop questioning."
>
> —— Albert Einstein

The motor ontology does not contradict existing physical theories. It gives them a foundation.

GR describes the curvature of spacetime. In our model, this is the result of the gravitational dimension acting at each step.

Quantum mechanics describes probabilities. In our model, quantum uncertainty is a consequence of our not knowing all the data in the current frame. Entanglement is explained by the synchronisation of frames.

The Standard Model remains valid inside a frame, but the motor dimension predicts new particles – motor copies.

The main point: physics describes the film script, the motor ontology describes the projector.

Chapter 11

Unanswered questions

"The more I learn, the more I realise how much I don't know."

—— Albert Einstein (attributed)

Any honest model must acknowledge its limits.

1. Where did the laws of physics come from? We do not know. 2. Where did the initial state come from? We do not know. 3. How exactly is consciousness linked to the motor dimension? We do not know. 4. Is there a connection between individual consciousnesses? We do not know. 5. What happens to consciousness after falling out of the flow ($n = 0$)? We do not know. 6. Are there other universes with different frequencies? We do not know.

But that is normal. Science has always moved from ignorance to knowledge.

Chapter 12

The human being in this picture of the world

> "We are not passive spectators, but active participants in the cosmic drama."
>
> —— Freeman Dyson (paraphrased)

We are not spectators, but co-authors. Every choice we make changes reality. This gives us responsibility and meaning.

We do not know how many frames we are given, but each one is valuable because it is ours.

The meaning of life lies in the very participation in the process. We build the world in which we want to live.

We are connected through a common flow, and our actions affect others.

The motor ontology is not a religion, but it opens a space for dialogue between science and spirituality.

Epilogue

> "The journey is the destination."
>
> —— Dan Eldon

This book does not put a full stop. It puts an ellipsis. Questions are deeper than answers.

I invite you, reader, to continue this path. Reality is a dialogue, and we are all its participants.